Polymers

Answer key with worked solutions

A. Understanding

1. Definitions.

> (a) **Monomer:** a small molecule that joins with many others to form a polymer. (b) **Repeat unit:** the part of the chain that repeats, drawn in brackets with n. (c) **Addition polymerization:** monomers with C=C join as the double bonds open; no other product. (d) **Condensation polymerization:** monomers with two functional groups join, releasing a small molecule (usually water) at every link.

B. Repeat units and monomers

2. Repeat units.

> (a) **–[CH₂–CH(CH₃)]–** (b) **–[CH₂–CHCl]–** (c) **–[CF₂–CF₂]–** (d) **–[CH₂–CH(C₆H₅)]–**, each with n after the brackets.

3. Monomers.

> (a) **propenenitrile** (acrylonitrile), $CH_2{=}CHCN$ (b) **propene**, $CH_2{=}CHCH_3$ (c) **1,1-dichloroethene**, $CH_2{=}CCl_2$

4. Classification.

> **Addition:** poly(ethene), PVC, PTFE. **Condensation:** nylon-6,6, PET, protein.

5. Nylon-6,6.

> (a) An **amide** link, –CO–NH– (a C=O bonded to an N–H), formed between –COOH and –NH₂. (b) **Water**, H₂O, one molecule per link.

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C. Calculations

6. Degree of polymerization.

> (a) $n = \dfrac{3.00 \times 10^{5}\ \text{g/mol}}{100.02\ \text{g/mol}} = \mathbf{3.00 \times 10^{3}}$ repeat units
>
> (b) $M = 3.00 \times 10^{3} \times 42.08\ \text{g/mol} = \mathbf{1.26 \times 10^{5}\ g/mol}$

7. Fluorine in PTFE.

> Repeat unit C_2F_4 has 4 F: $\dfrac{4 \times 19.00\ \text{g/mol}}{100.02\ \text{g/mol}} \times 100\ \% = \mathbf{76.0\ \%}$

8. Water from nylon.

> $n(\text{repeat units}) = \dfrac{1.00 \times 10^{3}\ \text{g}}{226.32\ \text{g/mol}} = 4.419\ \text{mol}$
>
> $n(\text{H}_2\text{O}) = 2 \times 4.419\ \text{mol} = 8.837\ \text{mol}$
>
> $m(\text{H}_2\text{O}) = 8.837\ \text{mol} \times 18.02\ \text{g/mol} = \mathbf{159\ g}$

D. Challenge

9. Burning poly(ethene).

> (a) $\text{C}_2\text{H}_4 + 3\,\text{O}_2 \longrightarrow 2\,\text{CO}_2 + 2\,\text{H}_2\text{O}$ (per repeat unit)
>
> (b) $n(\text{repeat units}) = \dfrac{1.00 \times 10^{3}\ \text{g}}{28.05\ \text{g/mol}} = 35.65\ \text{mol}$; $n(\text{CO}_2) = 2 \times 35.65\ \text{mol} = 71.30\ \text{mol}$
>
> $m(\text{CO}_2) = 71.30\ \text{mol} \times 44.01\ \text{g/mol} = 3.138 \times 10^{3}\ \text{g} = \mathbf{3.14\ kg}$
>
> (c) Advantage: **energy is recovered** and the waste volume is greatly reduced. Disadvantages: it releases **CO_2** (a greenhouse gas: over three times the mass of the plastic) and may release **toxic gases** (CO from incomplete combustion, HCl from PVC); the raw material is **lost**, whereas recycling saves crude oil and energy.

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